Programming Fundamentals Lab#4

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- Assignment#4
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All codes will be uploaded to my Github a few days after the submission of this assignment:

https://github.com/MF4rhan

Task # 1: Check Voting Eligibility

Write a program that asks the user for their age and prints whether they are eligible to vote or not.

- Use if-else statement.

```
#include <stdio.h>
int main() {
 int Age;
 printf("Enter your age to check for your eligibility to Vote.\n");
 scanf("%d", &Age);
 if (Age>=18)
 {
   printf("You are eligibile to Vote.");
 }
 else
 {
   printf("You are not eligibile to Vote.");
 }
 return 0;
}
```

```
Enter your age to check for your eligibility to Vote.
20
You are eligibile to Vote.
```

```
Enter your age to check for your eligibility to Vote.
18
You are eligibile to Vote.
```

```
Enter your age to check for your eligibility to Vote.
15
You are not eligibile to Vote.
```

Task # 2: Odd or Even Number

Write a program to check if a number entered by the user is odd or even.

- Use if statement with the modulus operator.

```
#include <stdio.h>
int main() {
 int Num;
 printf("Enter any number to check if its Even or Odd.\n");
 scanf("%d", &Num);
 if (Num%2==0)
 {
   printf("It is an Even Number.");
 }
 else
 {
   printf("It is an Odd Number.");
 }
 return 0;
}
```

```
Enter any number to check if its Even or Odd.
5
It is an Odd Number.
```

```
Enter any number to check if its Even or Odd.
6
It is an Even Number.
```

Task # 3: Find Largest Among Three Numbers

Write a program to find the largest number among three numbers input by the user.

- Use if-else if-else statements.

```
#include <stdio.h>

int main()

{

   int Num1,Num2,Num3;

   printf("Enter three numbers (separated by a space) to show the
largest.\n");

   scanf("%d %d %d",&Num1,&Num2,&Num3);

   if (Num1>Num2 && Num1>Num3)

   {

      printf("%d is the largest number.",Num1);

   }

   else if (Num2>Num1 && Num2>Num3)

   {

      printf("%d is the largest number.",Num2);

   }

   else if (Num3>Num1 && Num3>Num2)

   {

      printf("%d is the largest number.",Num3);

   }

   else

   {

      printf("All three numbers are equal.");
```

```
    }
    return 0;
}
```

```
Enter three numbers(seperated by a space) to show the largest.
89 56 74
89 is the largest number.
```

```
Enter three numbers(seperated by a space) to show the largest.
53 79 23
79 is the largest number.
```

```
Enter three numbers(seperated by a space) to show the largest.
24 65 96
96 is the largest number.
```

```
Enter three numbers(seperated by a space) to show the largest.
55 55 55
All three numbers are equal.
```

Task # 4: Grade Calculator

Create a program that takes a student's score as input and prints the grade based on the following:

- 90 and above: A
- 80 to 89: B
- 70 to 79: C
- 60 to 69: D
- Below 60: F
- Use if-else if-else structure.

```
#include <stdio.h>
int main()
{
    int Marks;
    printf("Enter your Marks.\n");
    scanf("%d",&Marks);
    if (Marks>=90)
    {
        printf("You have obtained A grade.");
    }
    else if (Marks>=80 && Marks<90)
    {
        printf("You have obtained B grade.");
    }
    else if (Marks>=70 && Marks<80)
    {
        printf("You have obtained C grade.");
```

```
    }
    else if (Marks>=60 && Marks<70)
    {
        printf("You have obtained D grade.");
    }
    else
    {
        printf("You have obtained F grade.");
    }
    return 0;
}
```

```
Enter your Marks.
92
You have obtained A grade.
```

```
Enter your Marks.
83
You have obtained B grade.
```

```
Enter your Marks.
76
You have obtained C grade.
```

```
Enter your Marks.
69
You have obtained D grade.
```

```
Enter your Marks.
57
You have obtained F grade.
```

Task # 5: Traffic Signal Control

Simulate a traffic light where the user inputs a color (1 for Red, 2 for Yellow, 3 for Green). The program prints the

action to be taken: “Stop”, “Ready” or “Go”.

- Use switch-case.

```
#include <stdio.h>
int main()
{
    int color;
    printf("Enter 1 for Red, 2 for Yellow, 3 for Green.\n");
    scanf("%d", &color);
    switch (color)
    {
    case 1:
        printf("Stop.\n");
        break;
    case 2:
        printf("Ready.\n");
        break;
    case 3:
        printf("Go.\n");
        break;
    default:
    printf("Invalid Input, Please enter 1 or 2 or 3.");
        break;
```

```
    }
    return 0;
}
```

```
Enter 1 for Red, 2 for Yellow, 3 for Green.
1
Stop.
```

```
Enter 1 for Red, 2 for Yellow, 3 for Green.
2
Ready.
```

```
Enter 1 for Red, 2 for Yellow, 3 for Green.
3
Go.
```

```
Enter 1 for Red, 2 for Yellow, 3 for Green.
4
Invalid Input, Please enter 1 or 2 or 3.
```

Task # 6: Basic Calculator

Write a program that takes two numbers and an operator (+, -, *, /) as input and performs the corresponding arithmetic operation.

- Use switch-case statement.

```
#include <stdio.h>

int main()

{

   int Op;

   float Num1, Num2;

   printf("Enter any two numbers (seperated by space).\n");

   scanf("%f %f", &Num1, &Num2);

   printf("Enter 1 for Addition, 2 for Subtraction, 3 for Product and 4 for
Division.\n");

   scanf("%d", &Op);

   switch (Op)

   {

   case 1:

      printf("The sum of %.2f and %.2f is %.2f.\n", Num1, Num2, Num1+Num2);

      break;

   case 2:

      printf("The difference of %.2f and %.2f is %.2f.\n", Num1, Num2, Num1-
Num2);

      break;

   case 3:

      printf("The product of %.2f and %.2f is %.2f.\n", Num1, Num2, Num1*Num2);

      break;
```

```
    case 4:
        printf("The quotient of %.2f and %.2f is %.2f.\n", Num1, Num2, Num1/Num2);
        break;
    default:
        printf("Invalid Operation Number, Please Enter 1, 2, 3 or 4 only.");
        break;
    }
    return 0;
}
```

```
Enter any two numbers (seperated by space).
7 5
Enter 1 for Addition, 2 for Subtraction, 3 for Product and 4 for Division.
1
The sum of 7.00 and 5.00 is 12.00.
```

```
Enter any two numbers (seperated by space).
7 5
Enter 1 for Addition, 2 for Subtraction, 3 for Product and 4 for Division.
2
The difference of 7.00 and 5.00 is 2.00.
```

```
Enter any two numbers (seperated by space).
7 5
Enter 1 for Addition, 2 for Subtraction, 3 for Product and 4 for Division.
3
The product of 7.00 and 5.00 is 35.00.
```

```
Enter any two numbers (seperated by space).
7 5
Enter 1 for Addition, 2 for Subtraction, 3 for Product and 4 for Division.
4
The quotient of 7.00 and 5.00 is 1.40.
```

```
Enter any two numbers (seperated by space).
7 5
Enter 1 for Addition, 2 for Subtraction, 3 for Product and 4 for Division.
5
Invalid Operation Number, Please Enter 1, 2, 3 or 4 only.
```

Task # 7: Leap Year Check

Create a program to check if a year entered by the user is a leap year or not.

- Use if-else with logical operators.

```
#include <stdio.h>
int main()
{
    int year;
    printf("Enter a Year to check if its a Leap Year.\n");
    scanf("%d", &year);
    if (year%400==0 || year%4==0 && year%100!=0)
    {
        printf("It is a Leap Year.");
    }
    else
    {
        printf("It is Not a Leap Year.");
    }
    return 0;
}
```

```
Enter a Year to check if its a Leap Year.
1999
It is Not a Leap Year.
```

```
Enter a Year to check if its a Leap Year.
2000
It is a Leap Year.
```

```
Enter a Year to check if its a Leap Year.
2004
It is a Leap Year.
```

```
Enter a Year to check if its a Leap Year.
2100
It is Not a Leap Year.
```

Task # 8: Number Sign Checker

Write a program to check if a number is positive, negative, or zero.

- Use if-else if-else statement.

```
#include <stdio.h>
int main()
{
   int Num;
   printf("Enter a number and I will check if it is positive, negative, or zero.\n");
   scanf("%d", &Num);
   if (Num>0)
   {
      printf("%d is a Positive Number.", Num);
   }
   else if (Num<0)
   {
      printf("%d is a Negative Number.", Num);
   }
   else
   {
      printf("The input number is Zero.");
   }
   return 0;
}
```

```
Enter a number and I will check if it is positive, negative, or zero.
5
5 is a Positive Number.
```

```
Enter a number and I will check if it is positive, negative, or zero.
0
The input number is Zero.
```

```
Enter a number and I will check if it is positive, negative, or zero.
-6
-6 is a Negative Number.
```

Task # 9: Day Name from Number

Write a program where the user enters a number (1-7), and the program prints the corresponding day of the week.

- Use switch-case.

```
#include <stdio.h>
int main()
{
    int Day;
    printf("Enter a number (1-7) to get the respective day of the week.\n");
    scanf("%d",&Day);
    switch (Day)
    {
    case 1:
        printf("Monday.\n");
        break;
    case 2:
        printf("Tuesday.\n");
        break;
    case 3:
        printf("Wednesday.\n");
        break;
    case 4:
        printf("Thursday.\n");
        break;
    case 5:
```

```
        printf("Friday.\n");
        break;
    case 6:
        printf("Saturday.\n");
        break;
    case 7:
        printf("Sunday.\n");
        break;
    default:
    printf("Invalid Input, Please enter a number 1 - 7.");
        break;
    }
    return 0;
}
```

```
Enter a number (1-7) to get the respective day of the week.
1
Monday.
```

```
Enter a number (1-7) to get the respective day of the week.
2
Tuesday.
```

```
Enter a number (1-7) to get the respective day of the week.
3
Wednesday.
```

```
Enter a number (1-7) to get the respective day of the week.
4
Thursday.
```

```
Enter a number (1-7) to get the respective day of the week.
5
Friday.
```

```
Enter a number (1-7) to get the respective day of the week.
6
Saturday.
```

```
Enter a number (1-7) to get the respective day of the week.
7
Sunday.
```

```
Enter a number (1-7) to get the respective day of the week.
8
Invalid Input, Please enter a number 1 - 7.
```

Task # 10. Simple Password Validator

Ask the user to enter a password (integer). If the password matches a predefined value, print “Access Granted” else print “Access Denied”, do it by using if-else.

```
#include <stdio.h>
int main()
{
   int Pass;
   printf("Enter your Password (Integers Only).\n");
   scanf("%d",&Pass);
   if (Pass==0000)  // 0000 could be any predefined value.
   {
      printf("Access Granted.");
   }
   else
   {
      printf("Acess Denied.");
   }
return 0;
}
```

```
Enter your Password (Integers Only).
0000
Access Granted.
```

```
Enter your Password (Integers Only).
1111
Acess Denied.
```